

Understanding System and User node pool types

- **System node pool**
 - Intended to be run crucial system pods like coredns, konnectivity or metrics-server
 - Its nodes get **kubernetes.azure.com/mode: system** label automatically added, making AKS prefer scheduling system pods on System node pool(s). Achievable because the system pods are configured with a nodeAffinity type **preferredDuringSchedulingIgnoredDuringExecution** to match that label
- **User node pool**
 - Should be used to host application pods
 - To prevent application pods from getting scheduled on System node pools, **CriticalAddonsOnly=true:NoSchedule** taint needs to be added at System node pool level. System pods are configured with a toleration for that taint, but the application pods not
- It is not mandatory to separate the workload between a system and a user node pool. You need to have at least one system node pool, and it's ok if it hosts both, system and application pods, but it's recommended to split system and application pods if possible
- You can convert the type of a node pool (from System to User or vice-versa), if there is at least one System node pool in the cluster

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Consideration/ Limitation	System	User
OS type	Must be Linux	Can be Linux or Windows
Minimum node pools in the cluster	At least one System node pool must exist within the cluster	Having a User node pool is not mandatory
Minimum node pool count	At least one node	User node pool can have 0 nodes or more
Stop particular node pool	Cannot be stopped	Can be stopped
MaxPods interval	30 - 250	10 -250
Spot node pools supportability	Not supported	Supported